The Assay That Lies

Every batch of a hemoglobin product is released on the strength of a test built from the blood of a horseshoe crab, and the product itself is one of the more awkward things that test has ever been asked to measure

Before a single vial of a BHOC reaches a patient, somebody has to say that it is free of bacterial endotoxin, and the person saying it is relying on a clotting reaction discovered in the blood of an animal that has changed very little in hundreds of millions of years. The arrangement is so old and so routine that it has almost vanished from the conversation about oxygen carriers. Papers discuss vasoconstriction, nitric oxide scavenging, methemoglobin, and the endpoints of trials. They rarely discuss the humble release assay, even though the assay is the gate that every lot must pass and even though hemoglobin is, for reasons of plain physical chemistry, a difficult thing to pass through it.

This essay is about that gate. It is a story about measurement, about what a number on a certificate of analysis can and cannot promise, and about an ethical wrinkle that the field has hardly noticed: the safety of a human product depends on the periodic bleeding of a wild marine animal. I will say plainly where my sources run thin. Several of the most relevant papers sit behind publisher walls that I could not open while preparing this, so where I know a paper only by its title, I say so, and where I reason from general principles rather than from a measured result, I say that too.

A gel in a crab

The story begins in the 1950s. Fred Bang, an American medical researcher, observed in 1956 that bacteria of the gram-negative kind cause the blood of the horseshoe crab to gel. The animal has no adaptive immune system of the kind a mammal has, and its circulating cells, called amebocytes, carry granules loaded with a clotting protein named coagulogen. When endotoxin, the lipopolysaccharide that sits in the outer wall of gram-negative bacteria, touches the system, a sensor enzyme called factor C is switched on, and that begins a cascade in which one enzyme activates the next until coagulogen is converted and the sample sets into a gel. In the crab this walls off an infection. In a laboratory it announces that a contaminant is present.

The United States Food and Drug Administration approved the lysate test for drugs and devices in 1977, and in doing so it displaced the rabbit pyrogen test, in which a product was injected into rabbits and their body temperature watched for a fever. The lysate method was faster, cheaper, and far more sensitive, and it did not require a colony of rabbits. It became the standard way to ask whether a parenteral product carried a dangerous load of bacterial debris, and in its modern forms it still is.

The reason endotoxin matters so much for a product injected into the bloodstream is that the human response to it is out of all proportion to its mass. Minute quantities provoke fever, and larger exposures drive the inflammatory cascade that, at its extreme, produces shock. Any intravenous product has an endotoxin limit, set in units rather than in grams because the

potency of the stuff varies. I will not quote numerical limits for hemoglobin products, since I could not verify them.

Why hemoglobin is a bad subject for this test

Here the trouble begins. The lysate assay is a chemical reaction read by eye, by turbidity, or by color, and hemoglobin is a vividly colored protein that absorbs light strongly in exactly the part of the spectrum where optical readings are taken. A chromogenic endotoxin assay releases a yellow dye and reads its absorbance. A sample that is itself deeply red, and whose color shifts as the iron changes oxidation state, is not a neutral background. Even without the reasoning about color, the general framework used by regulators acknowledges that products can be inhibitory, lowering the apparent endotoxin below what is truly present, or enhancing, raising it above. The regulator therefore insists that a manufacturer demonstrate that endotoxin added to the product can actually be recovered by the assay before the assay is allowed to release product.

Color is only the first problem and probably the more tractable one, since a sample can be diluted until its color fades below the threshold that matters, provided the product contains little enough endotoxin that dilution does not push the signal beneath the detection limit. The second problem is chemical, and it is stranger. There is a body of literature, which I know here only by titles that came back in my searches, describing hemoglobin as an endotoxin binding protein, and one paper title states that hemoglobin binds lipopolysaccharide and enhances its biological activity. Another title, older, speaks of synergistic toxicity of endotoxin and hemoglobin. I did not read the abstracts, so I cannot tell you what doses, what animals, or what size of effect. What the titles imply, however, and what makes the idea worth holding in mind, is that hemoglobin and endotoxin do not simply coexist in a vial. They may interact, and the interaction may do two things at once. It may mask endotoxin from an assay that expects it free in solution, and it may make a given amount of endotoxin hit harder in a living body.

Consider what that would mean if both effects were real in a given product. The assay would read low, because bound lipopolysaccharide is partly hidden from the lysate cascade. The patient would respond high, because the bound complex is more provocative than the free molecule. The certificate would say the lot was clean at the same moment that biology said otherwise. That is the sense in which the assay can lie, and I use the word advisedly: it would not be lying through any fault, only by answering a different question from the one that the clinician needs answered. The assay asks how much endotoxin the lysate can detect under the conditions of the test. The clinician needs to know how much inflammatory stimulus the vial will deliver to a human being. For most products those two questions have nearly the same answer. For a protein that binds the analyte, they may not.

What the early solutions may have taught

The first hemoglobin solutions given to people were made by lysing red cells and removing the membranes, and the products were called stroma-free hemoglobin. Their history in the twentieth century is one of enthusiasm followed by disappointment, with kidney injury and other toxicities attributed to them. For a long time the explanation offered was that the stroma had not been completely removed, or that the dimers of hemoglobin damaged the kidney. At some point researchers began asking whether endotoxin had been an unrecognized passenger. A 1993 paper in the journal Transfusion, whose title I saw but whose text I could not open,

concerns the role of quantitative endotoxin determination in producing modified cross-linked cell-free hemoglobin for human use. The mere existence of such a paper tells us that by the early nineties the manufacturers of these products regarded endotoxin measurement as a central production concern rather than a formality.

I will not claim that endotoxin explains the toxicity of the early solutions. I do not have the evidence for that, and to my knowledge the field has not reached a consensus that it does. What can be said is logical rather than historical. If hemoglobin potentiates endotoxin, then an early product contaminated with amounts that would have been tolerable alone might have produced fever, vascular effects, or organ stress that seemed intrinsic to the hemoglobin. Researchers then, working with assays whose behavior in hemoglobin was not fully characterized, could not have excluded that possibility, and neither can we, looking backward. The honest statement is that some portion of the early record is uninterpretable on this one axis, and that this is an uncomfortable thing to say about a body of work that trained a generation of investigators.

The bovine twist

A bovine product adds an extra consideration that the human-derived products did not face in the same way. The raw material is the blood of cattle, collected at a slaughterhouse or a controlled facility, and the animal is a ruminant whose gut carries an enormous burden of gram-negative bacteria. Blood collection is a clean process in principle, but it is not a sterile surgical one, and the first step of manufacture, breaking open the red cells, releases hemoglobin into a solution that is, from that moment, vulnerable to every endotoxin molecule that came in on the equipment, the water, the buffers, and the chromatography media. Endotoxin is notoriously sticky and notoriously heat stable. It survives ordinary sterilization, because killing the bacterium does not destroy the lipopolysaccharide in its wall. It adheres to plastic and glass and to the resins used in purification. The classic remedies are rigorous water quality, caustic washes of columns and tubing, and filtration steps designed to bind it, though I note these are general practices of biologics manufacture and I have not confirmed which of them any particular oxygen carrier maker uses.

What distinguishes a protein that carries a heme group from a monoclonal antibody, for example, is that the protein purification itself is hard on endotoxin removal. Many standard removal methods work by capturing the lipopolysaccharide on a charged surface, and a highly charged protein at high concentration competes for that surface or drags endotoxin along with it. A manufacturer therefore faces a conflict between making the product concentrated enough to carry useful oxygen, since the entire point is a high hemoglobin content per volume, and keeping it clean. This is a general expectation drawn from protein chemistry, and I flag it as reasoning, not as a documented finding about a named product.

The animal on the other end of the assay

Now to the ethical wrinkle. The lysate used in the standard test comes from horseshoe crabs, which are caught, bled, and in most operations returned to the water. The extraction takes a significant portion of the animal’s blood. The question of what happens to the crab afterward has been argued for years. Industry statements before 2008 held that mortality after bleeding was below three percent. A study in Massachusetts reported in 2009 found higher figures, with

mortality of roughly five to fifteen percent for males and estimates reaching about twenty-nine percent for females, and the same study found differences according to whether females were returned immediately or kept overnight, with roughly twenty-two percent mortality for the first and thirty percent for the second. These are the numbers as reported in the source I could read, a secondary summary, and I treat them as approximate and as belonging to one region and one set of handling practices. The mortality in other places and under other practices is not something I can state.

What interests me is the moral structure rather than the crab arithmetic. A life-saving product for humans leans on an unpaid and unconsenting partner, and the lean is invisible because it sits in a quality control laboratory far from the bedside. If a BHOC ever becomes a routine product in a trauma bay, the number of lots tested, and therefore the number of crabs bled, becomes part of its true cost, and an honest account of the product would include it. I do not think the quantity would be large next to the existing pharmaceutical industry's use. I cannot give a number, and I decline to guess. The point is that the cost exists and is currently counted nowhere in the discussion of oxygen carriers.

A synthetic reagent arrives

There is an alternative that bears directly on our subject. Recombinant factor C, usually abbreviated rFC, reproduces the first enzyme of the cascade using genetically modified insect cells, and so requires no crab. It has been available since 2003. The American and European regulators acknowledged it as acceptable in 2012, the European Pharmacopoeia included it in 2016, and the United States Pharmacopeia gave it official standing as of May 2025, which one summary from the Marine Biological Laboratory described as leveling the playing field between crab blood and the synthetic alternatives.

For a maker of a hemoglobin product this opens a door and a question at once. The door is obvious: no crab dependence, a defined reagent without lot variability from a wild animal, and a clean ethical profile. The question is the one this whole essay has circled. Does the synthetic test behave better or worse in the presence of a heme protein? The rFC assay uses a fluorescent readout rather than a clot or a chromogenic dye, and fluorescence has its own vulnerabilities, since a strongly absorbing colored sample can quench the signal, an effect known as the inner filter. Whether that is an advantage depends on the wavelengths involved, and I have not seen a head to head comparison in hemoglobin matrices. My expectation, which is only an expectation, is that neither method will be uniformly better, that each will fail in a different way, and that the sensible course is to run both on development lots and to compare them against a spike recovery panel designed for the product.

This matters because the old pattern was to adopt whichever method the regulator had always accepted and to treat the question as closed. The 2025 shift removes the excuse. For the first time a developer can ask which test is truthful for this protein, rather than which test is customary, and can generate the comparative data that will be the first of its kind in the literature if anyone bothers to publish it.

What a better release test would look like

If I were designing the release program for an oxygen carrier from scratch, I would begin by refusing to treat endotoxin as a single number. I would want at least three kinds of information for every development lot. The first is the assay reading at several dilutions, so that any trend with dilution, the signature of interference, becomes visible. A product that reads the same endotoxin level at ten fold and at one hundred fold dilution is behaving well, and a product whose apparent level climbs as it is diluted is revealing that something in the undiluted matrix was suppressing the signal. The second is the spike recovery, with a spike introduced early, before the final purification step, as well as at the end, to catch the possibility that processed endotoxin is hidden in ways that a fresh spike is not. The third is a biological readout that does not use a lysate at all, such as a monocyte activation test, in which human immune cells respond to the sample with cytokine release. Such a test measures what the human system does, not what a clotting enzyme does, and it responds to inflammatory contaminants other than endotoxin. It is slower and more variable, but it asks the clinician's question. Its use as an alternative to the rabbit test is established in some regulatory settings, though I would not want to overstate which, and its behavior in a heme protein matrix would itself need validation, since red color and iron could perturb the cell readout in their own way.

Where the gate leads

A few beliefs, each marked with how firmly I hold it.

I believe, with confidence, that a lysate assay has to be validated in every new matrix, and that a colored, protein-dense matrix is among the hardest cases there is. This follows directly from how the test works and from the regulatory demand for recovery proof.

I believe, with moderate confidence, that the interaction between hemoglobin and endotoxin is real and biologically relevant, because the literature titles I found point that way and because the physical chemistry of a sticky amphipathic molecule meeting a heme protein makes it plausible. I could not read the primary papers, and a reader who cares should go to them before leaning on my paraphrase.

And I believe, with fair confidence, that the 2025 pharmacopoeial acceptance of recombinant factor C changes the position of every developer of a blood-derived biologic. It removes a dependence on a wild animal, and it creates the opportunity for the first honest comparison of test methods in a heme matrix. A developer who takes that opportunity will learn something about their own product that competitors are not bothering to learn.

There is a deeper moral to the story, and it is a modest one. Measurement in medicine is always a proxy, and the proxies we trust most are the ones we have stopped examining. The endotoxin test is old, trusted, and tied to a precise legal definition of clean. It is also built from the immune machinery of an animal that evolved it for a different purpose, applied to a protein that interferes with it, reported as a single number to a clinician who reads that number as a guarantee. A field that wants to put an oxygen carrier into a bleeding human being should be able to say, without flinching, what the number means and where it stops meaning anything. At present, in the parts of the literature I could examine, that statement is rarely made. It would be a small and inexpensive act of seriousness to begin making it.

Further reading

The history of the lysate test is summarized in general references on Limulus amebocyte lysate, including its origin in the 1956 observation by Fred Bang, its regulatory adoption in 1977, and its recombinant successors. The Marine Biological Laboratory published a short account of the United States Pharmacopeia decision of 2025 on synthetic alternatives. Fisheries management documents from the Atlantic States Marine Fisheries Commission describe the horseshoe crab harvest. For hemoglobin specifically, readers should look for the 1993 Transfusion paper on quantitative endotoxin determination in modified cross-linked cell-free hemoglobin, the chapter on measurement of endotoxin levels in hemoglobin preparations in a Methods in Enzymology volume, and the papers in Artificial Cells, Blood Substitutes, and Biotechnology on hemoglobin as a lipopolysaccharide binding protein and on synergistic toxicity of endotoxin and hemoglobin. I knew these only by title or abstract fragments, so verify their contents before relying on my description.